HCIC Non-Python Study Guide

Tier 2 Topics • Appears Most Years (2–3 of 4 papers)

This guide covers the seven non-Python topics that have appeared in most HCIC papers. Study these once the Tier 1 topics are solid — they are the difference between a good and a great score.

The Seven Tier 2 Topics

1. AI / Machine Learning / Deep Learning — concepts and hierarchy
2. Gear Rotation Direction — which way each gear spins
3. Omni-wheels and Mecanum Wheels — omnidirectional movement
4. Sensor Configuration — placing sensors for optimal line following
5. Maze Navigation Algorithms — how robots solve mazes
6. Microcontroller / Arduino Code — reading basic C++ sketches
7. Logic Gates and Truth Tables — AND, OR, NOT, NOR

Topic 1 — Artificial Intelligence, Machine Learning and Deep Learning

HCIC tests your understanding of the relationship between AI, ML, and DL, and which statements about each are true or false. This is a conceptual topic — no calculations required.

The Hierarchy

The Key Relationship

Deep Learning $\subset$ Machine Learning $\subset$ Artificial Intelligence

DL is a subset of ML. ML is a subset of AI. AI is the broadest field.

What Each Term Means

Term	Definition
Artificial Intelligence (AI)	Building machines that can perform tasks that normally require human intelligence — reasoning, perception, decision-making.
Machine Learning (ML)	A subset of AI. Machines learn from data and experience rather than being explicitly programmed with rules.
Deep Learning (DL)	A subset of ML. Uses artificial neural networks with many layers to extract features from raw data (images, sound, text).

True Statements to Know

- AI aims to build machines capable of thinking like humans.
- ML is the practice of getting machines to make decisions through data, actions and experience.
- DL uses Artificial Neural Networks (ANNs) to solve complex problems.
- DL uses multiple layers to progressively extract higher-level features from raw input.
- Using ML, a robot can train itself by processing millions of images.
- DL is a subset of ML. ML is a subset of AI.

False Statements to Recognise

- **FALSE:** 'AI is a subset of ML' — it is the other way around. ML is a subset of AI.
- **FALSE:** 'Numerical password entry in a door lock is a breakthrough in AI' — this is basic digital input, not AI.
- **FALSE:** 'DL is a subset of AI but not ML' — DL is a subset of both ML and AI.

Worked Example

Which statements about AI/ML/DL are FALSE?

i	AI aims to build machines which are capable of thinking like humans.
ii	AI is a subset of Data Science.
iii	AI is a subset of ML which is a subset of DL.
iv	ML is the practice of getting machines to make decisions through data.
v	Using ML, the robot trains itself by processing millions of images.
vi	ML learns from data, actions and experience.
vii	DL is the process of using Artificial Neutral Networks to solve complex problems.
viii	DL is a subset of ML.
ix	DL uses multiple layers to progressively extract higher-level features from the raw input.
x	One breakthrough in AI includes numerical password entry in door unlocking or phone unlocking in the smart phone.

Answer

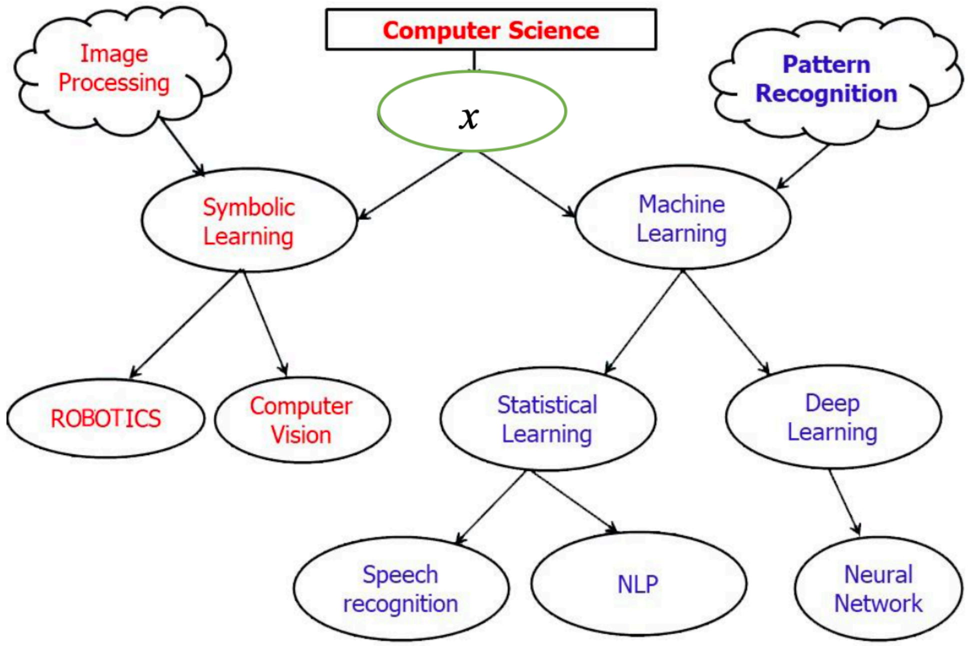
Statement iii: 'AI is a subset of ML which is a subset of DL' — FALSE (hierarchy is reversed)

Statement x: 'Numerical password entry is a breakthrough in AI' — FALSE (it's basic input, not AI)

Answer: (C) Only statements iii and x are false.

Worked Example

What goes in the X?



Answer

X sits between Computer Science and ML/Symbolic Learning.

It encompasses image processing, pattern recognition, robotics, computer vision, NLP, neural networks.

Answer: Artificial Intelligence.

Topic 2 — Omni-wheels and Mecanum Wheels

Mecanum and omni-wheels allow a robot to move in any direction (including sideways and diagonally) without rotating first. Each wheel has small rollers set at an angle (usually 45°) around its rim. Questions ask which wheel spin directions produce a given movement.

How They Work

- Each wheel drives in its axle direction, but the angled rollers produce a diagonal force.
- By combining wheel speeds and directions, the net forces add up to move the robot in any direction.
- A standard 4-wheel mecanum robot has wheels at each corner. The key is which wheels spin forward and which spin backward.

Movement Patterns for a 4-Wheel Mecanum Robot

For a robot with wheels at corners (Front-Left FL, Front-Right FR, Back-Left BL, Back-Right BR):

Movement	Wheels spinning FORWARD	Wheels spinning BACKWARD
Move Forward	FL, FR, BL, BR all forward	—
Move Backward	—	FL, FR, BL, BR all backward
Slide Right	FL, BR forward	FR, BL backward
Slide Left	FR, BL forward	FL, BR backward
Rotate Clockwise	FL, BL forward	FR, BR backward
Rotate Counter-CW	FR, BR forward	FL, BL backward

Worked Example — HCIC 2024 Q6 (Moving Around a Bend)

Given the mecanum wheel diagrams for 'Moving Forward' (all wheels same direction) and 'Moving Sideways' (alternate wheels opposite), which is the wheel directions for moving around a bend?

Answer

Moving around a bend = forward motion + rotation combined.

One side of wheels spins faster/forward while the other does nothing

This creates a turning force while the robot continues moving.

Answer: One side forward, other side not moving.

Topic 3 — Maze Navigation Algorithms

Maze-solving robots use a combination of sensors, memory, and search algorithms to navigate from a start point to an exit. HCIC tests which techniques are valid and how robots implement them.

Key Techniques

Technique	How it works
Wall-following (right-hand rule)	Always keep one hand (sensor) on the wall. Works for any maze connected to the exit.
Mental map / Visited tracking	Remember which paths have been taken to avoid revisiting dead ends.
Landmark recognition	Use distinctive features in the maze to know your location.
Depth-First Search (DFS)	Explore as far as possible down one path before backtracking. Systematic, guaranteed to find exit.
Breadth-First Search (BFS)	Explore all paths at the same depth before going deeper. Finds the SHORTEST path.
Machine learning / optimisation	Learn from previous attempts to find more efficient routes over time.

What a Maze-Solving Robot Actually Does

- Uses sensors (ultrasonic, infrared) to detect walls and open paths.
- Maps the maze — records which cells/paths have been visited.
- Applies a search algorithm (DFS or BFS) to decide which direction to take.
- May use machine learning to optimise the route after multiple attempts.

Worked Example — HCIC 2023 Q7 (Open Maze)

An open maze is shown with the escape route. Which techniques can be used to navigate it?

Evaluating each option

(i) Place right hand on wall — VALID for most mazes.

(ii) Create a mental map — VALID, helps avoid revisiting dead ends.

(iii) Look out for landmarks — VALID, helps with orientation.

(iv) Use depth-first search — VALID, systematic algorithm.

Answer: (D) All four techniques (i, ii, iii, iv) are valid.

Worked Example — HCIC 2024 Q8 (True/False Statements)

Which statements about maze-solving robots are correct?

Evaluating each option

(i) Maze-solving robots use sensors and algorithms like DFS/BFS — TRUE.

(ii) They rely SOLELY on trial and error — FALSE (they use systematic algorithms).

(iii) They use ML to learn from previous attempts — TRUE.

(iv) They follow a predetermined path regardless of layout — FALSE.

Answer: (B) Statements i and iii are correct.

DFS vs BFS — Key Difference

Algorithm	Behaviour	Finds shortest path?
Depth-First Search (DFS)	Goes deep down one path before backtracking	Not guaranteed
Breadth-First Search (BFS)	Explores all paths at same depth simultaneously	Yes — always finds shortest

Topic 4 — Microcontroller / Arduino Code

HCIC uses Arduino (C++) for microcontroller questions. You don't need to write Arduino code from scratch — you need to read it and understand what each part does.

Arduino Program Structure

Two essential functions

```
void setup()  { ... }
```

Runs ONCE when the Arduino powers on.

Used to configure pins, start serial communication, etc.

```
void loop()  { ... }
```

Runs REPEATEDLY, forever, after setup() finishes.

This is where the main robot behaviour goes.

Key Functions to Know

Function	What it does	Example
pinMode(pin, mode)	Sets a pin as INPUT or OUTPUT	pinMode(13, OUTPUT)
digitalWrite(pin, val)	Sets a pin HIGH or LOW (on/off)	digitalWrite(13, HIGH)
analogRead(pin)	Reads an analogue value (0–1023) from a pin	val = analogRead(A0)
analogWrite(pin, val)	Writes a PWM value (0–255) to a pin	analogWrite(9, 128)
delay(ms)	Pauses the program for a number of milliseconds	delay(1000) // 1 second
Serial.begin(baud)	Opens serial communication	Serial.begin(9600)
Serial.print(val)	Prints a value to the serial monitor	Serial.print(val)

Worked Example — HCIC 2022 A4 (pinMode)

What is the function of this code?

Code

```
void setup() {
   pinMode(kPinLed, OUTPUT);
}
```

Answer

This sets up the pin that the LED is connected to as an OUTPUT pin.

The setup() function runs once at the start.

pinMode with OUTPUT means the Arduino will send current through this pin (to light the LED).

Answer: (A) This sets up the pin that the LED is connected to as an OUTPUT pin.

Worked Example — HCIC 2022 A5 (while loop in Arduino)

Based on this Arduino sketch, which statement is FALSE?

Code

```
int a = 0;
while ( a < 10 ) {
    a = a + 1;
    delay(1000);
}
```

Evaluating each option

(A) It starts with variable a setting to 0 — TRUE.

(B) It adds 1 to the value of a and waits 1 second — TRUE.

(C) It repeats the process until a has a value of 10 — TRUE.

(D) Once a is equal to 10, the comparison in the while statement is true — FALSE.

When a=10, the condition 'a < 10' is FALSE, which is why the loop ends.

Answer: (D)

Worked Example — HCIC 2024 Q10 (Search in Arduino)

What objective does this Arduino loop() program serve?

Code

```
void loop() {
    int a[4] = {1, 2, 3, 4};
    int search_item = 2;
    for (i=0; i<4; i++) {
        if (a[i] == search_item) {
            Serial.print('Found!');
        }
    }
}
```

Answer

The program loops through an array looking for a specific value (search_item = 2).

When it finds a match, it prints 'Found!'.

Answer: (A) Search.

Topic 5 — Logic Gates and Truth Tables

Logic gates are the building blocks of digital circuits. They take binary inputs (0 or 1) and produce a binary output. HCIC tests whether you can identify a gate from its truth table, or fill in a truth table for a given gate.

The Four Gates You Need

Gate	Rule — when does output = 1?
AND	Only when BOTH inputs are 1
OR	When AT LEAST ONE input is 1
NOT	Flips the input — if input is 0, output is 1; if input is 1, output is 0
NOR	Opposite of OR — output is 1 only when BOTH inputs are 0

Truth Tables

Two-input gates (AND, OR, NOR):

A	B		AND		OR		NOR		NAND	Memory tip
1	1	→	1	→	1	→	0	→	0	Both 1: AND=1, OR=1, NOR=0
1	0	→	0	→	1	→	0	→	1	One 0: AND=0, OR=1, NOR=0
0	1	→	0	→	1	→	0	→	1	One 0: AND=0, OR=1, NOR=0
0	0	→	0	→	0	→	1	→	1	Both 0: AND=0, OR=0, NOR=1

How to Identify a Gate from its Truth Table

- **If output is 1 ONLY when both inputs are 1:** AND gate.
- **If output is 1 when at least one input is 1:** OR gate.
- **If output is 1 ONLY when both inputs are 0:** NOR gate.
- **If output is 0 ONLY when both inputs are 1:** NAND gate.
- **Single input, output is always the opposite:** NOT gate.

Worked Example — HCIC 2023 Q4

A truth table shows: 0,0→0 / 0,1→1 / 1,0→1 / 1,1→1. What logic type is this?

Answer

Output is 1 whenever at least one input is 1.

Output is only 0 when BOTH inputs are 0.

This is the OR gate.

Answer: (A) OR.

Worked Example — HCIC 2023 Q5

Which truth table shows the logic function of an AND gate?

Answer

AND gate: output is 1 ONLY when BOTH inputs are 1.

Truth table: 1,1→1 / 1,0→0 / 0,1→0 / 0,0→0.

Answer: (B) — the table where only the 1,1 row gives output 1.

Quick Revision Summary

AI / ML / DL Hierarchy

DL ⊂ ML ⊂ AI. AI is the broadest. DL is the narrowest.

AI = human-like intelligence. ML = learns from data. DL = neural networks.

AI is NOT a subset of Data Science. Numerical password entry is NOT AI.

Omni-wheels / Mecanum Wheels

Allow movement in ANY direction without turning the robot.

Forward: all wheels same direction. Sideways: diagonal-pair wheels oppose.

Turning: one side forward, other side backward.

Maze Navigation

Valid techniques: wall-following, mental map, landmarks, DFS, BFS, ML.

DFS = goes deep before backtracking. BFS = finds shortest path.

Robots use sensors to map, then algorithms to decide direction.

Arduino Code

setup() runs ONCE. loop() runs FOREVER.

pinMode(pin, OUTPUT/INPUT). digitalWrite(pin, HIGH/LOW). delay(ms).

analogRead() reads 0–1023. Serial.print() sends to monitor.

Logic Gates

AND: output 1 only if BOTH inputs are 1.

OR: output 1 if AT LEAST ONE input is 1.

NOR: output 1 only if BOTH inputs are 0 (opposite of OR).

NOT: single input — flips 0→1 and 1→0.

Practice Questions

AI / ML / DL

Q1. Which of the following correctly describes the relationship between AI, ML, and DL?
(A) AI ⊂ ML ⊂ DL (B) DL ⊂ ML ⊂ AI (C) ML ⊂ DL ⊂ AI (D) DL ⊂ AI ⊂ ML

Q2. True or False: 'Using ML, a robot can train itself by processing millions of images.'

Q3. True or False: 'AI is a subset of Data Science.'

Omni-wheels

Q4. A mecanum-wheel robot needs to slide directly sideways to the right. For a standard 4-corner layout, which wheels should spin forward and which backward?

Q5. What is the main advantage of mecanum wheels over regular wheels?

Maze Navigation

Q6. A robot uses the right-hand wall-following rule to solve a maze. The maze has an island in the centre (a wall not connected to the outer boundary). Will wall-following always find the exit? Explain.

Q7. What is the difference between Depth-First Search and Breadth-First Search in maze solving?

Arduino Code

Q8. What is the difference between setup() and loop() in an Arduino sketch?

Q9. What does delay(500) do in an Arduino program?

Logic Gates

Q10. Fill in the output column for an AND gate:
A=1, B=0 → ? A=0, B=0 → ? A=1, B=1 → ? A=0, B=1 → ?

Q11. A truth table shows output = 1 ONLY when both A=0 and B=0. Which gate is this?

Practice Question Answers

AI / ML / DL Answers

Q1. (B) DL ⊂ ML ⊂ AI. Deep Learning is the narrowest subset, AI is the broadest.

Q2. TRUE. This is a correct description of supervised machine learning with image data.

Q3. FALSE. AI is not a subset of Data Science — AI is an independent field. Data Science uses some AI tools, but AI is not contained within Data Science.

Omni-wheel Answers

Q4. For sliding RIGHT: Front-Left and Back-Right spin forward. Front-Right and Back-Left spin backward. The diagonal pairs oppose each other to produce sideways movement.

Q5. Mecanum wheels allow movement in any direction (including sideways and diagonally) without the robot needing to rotate first. Standard wheels can only move forward, backward, or turn by rotating in place.

Maze Navigation Answers

Q6. Not always. Wall-following only works reliably when all walls are connected to the outer boundary. If the maze has an island (isolated inner wall), the robot may loop around the island forever without finding the exit.

Q7. DFS goes as deep as possible along one path before backtracking — it may find a solution but not necessarily the shortest. BFS explores all paths at the same depth simultaneously — it always finds the shortest path to the exit.

Arduino Code Answers

Q8. setup() runs ONCE when the Arduino powers on — used for one-time configuration (e.g. setting pin modes). loop() runs repeatedly and forever after setup() finishes — it contains the main robot behaviour.

Q9. delay(500) pauses the program for 500 milliseconds (0.5 seconds). The Arduino does nothing during this time.

Logic Gates Answers

Q10. AND gate: A=1,B=0→0 / A=0,B=0→0 / A=1,B=1→1 / A=0,B=1→0. AND outputs 1 ONLY when both inputs are 1.

Q11. NOR gate. NOR outputs 1 only when BOTH inputs are 0 — the exact opposite of OR.

End of Tier 2 Non-Python Study Guide